Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

BCSE302L – DATABASE SYSTEMS LAB

DIGITAL ASSIGNMENT -1

Project Title: MaterCare: Pregnancy & Newborn Care Management System

A2+TA2

Submitted By:

1) Arya Mishra (24BCE1220)
2) Sakshi Pal (24BCE1316)

Problem Statement

Pregnancy and early newborn care require continuous monitoring of medical appointments, diagnostic tests, medications, and vaccination schedules. In real-world scenarios, this information is often scattered across multiple sources such as handwritten notes, hospital records, messaging applications, and disconnected digital platforms. Most existing pregnancy applications primarily focus on providing educational or informational content but lack structured data storage, tracking, and management capabilities.

This fragmented approach makes it difficult for expectant mothers and new parents to maintain accurate records, adhere to medical schedules consistently, and retrieve historical information when needed. Therefore, there is a need for a centralized, database-driven system that efficiently organizes pregnancy and newborn-related data while ensuring data integrity, security, and scalability.

MaterCare aims to address this problem by providing a unified platform that enables users to systematically manage pregnancy and newborn care information through a well-designed relational database and backend system, without offering medical diagnosis or treatment.

Objectives

- To design a comprehensive relational database for managing pregnancy and newborn care data.
- To centralize user information, pregnancy profiles, baby records, and emergency contacts in a structured format.
- To manage doctor, hospital, and appointment scheduling efficiently using proper entity relationships.
- To track medical tests and store test records linked to pregnancy profiles.
- To maintain medication records including dosage and treatment duration.
- To manage newborn vaccination schedules and track vaccination status.
- To implement reminder functionality for appointments, medications, and vaccinations.
- To ensure data integrity using primary keys, foreign keys, and well-defined relationships.
- To maintain system transparency and accountability through audit logs.
- To design a scalable backend database structure suitable for future enhancements.

Scope

- User registration, login, and secure authentication.

- Creation and management of pregnancy profiles.
- Management of doctor, hospital, and availability information.
- Appointment booking and status tracking.
- Recording and managing medical tests and test results.
- Tracking prescribed medications and treatment duration.
- Creation of baby profiles after childbirth.
- Managing vaccination schedules and vaccination records.
- Reminder generation for appointments, medications, and vaccinations.
- Maintenance of audit logs for system activity tracking.
- Implementation of a normalized relational database with referential integrity.

Assumptions of the Project

- Each pregnancy profile and baby profile belongs to exactly one registered user.
- A user may exist without pregnancies, babies, or reminders initially.
- Every appointment is linked to one pregnancy profile, one doctor, and one hospital.
- Doctors are assumed to be associated with only one hospital for simplicity.
- Master tables such as MedicalTest, Medication, and Vaccination are predefined and reused across records.
- Pregnancy profiles and baby profiles may exist without medical records or vaccination entries at the beginning.
- Reminders can be linked to a pregnancy profile, a baby profile, or only a user.
- Uploaded documents are stored as file references; actual file storage is outside the database scope.
- Audit logs are automatically created and are not modified after insertion.
- The system is designed for data tracking and management only, not for medical diagnosis or treatment.

Modules of the System

1. User Management Module

This module manages user registration, authentication, and profile maintenance. It allows users to create accounts, log in securely, update personal information, and manage emergency contact details. The module ensures secure access and personalized handling of user data.

2. Pregnancy Management Module

This module enables users to create and manage pregnancy profiles. It records essential details such as pregnancy start date and expected due date. All pregnancy-related activities, including appointments, medical tests, medications, documents, and reminders, are linked to this module. It functions as the core component of the system.

3. Doctor and Hospital Management Module

This module maintains information related to doctors and hospitals. It includes doctor details such as name, specialization, and contact information, as well as hospital information. It also manages doctor availability schedules and the employment relationship between hospitals and doctors.

This module allows users to schedule and manage medical appointments. It links pregnancy profiles with doctors and hospitals. The module tracks appointment details such as appointment ID, doctor, hospital, and status.

5. Medical Test Management Module

This module manages medical test records during pregnancy. It stores details of medical tests and records test results along with test dates. It maintains the relationship between pregnancy profiles and medical tests through structured records.

6. Medication Management Module

This module manages prescribed medications during pregnancy. It stores medication details, dosage information, and medication duration (start and end dates). It maintains proper linkage between pregnancy profiles and medication records.

7. Baby and Vaccination Management Module

This module handles baby profile creation after delivery. It records baby details such as name, date of birth, and gender. It also manages vaccination schedules and vaccination records, including vaccination dates and status.

8. Reminder Management Module

This module generates and manages reminders related to pregnancy, baby care, appointments, and vaccinations. It tracks reminder type, due date, and status to ensure timely notifications.

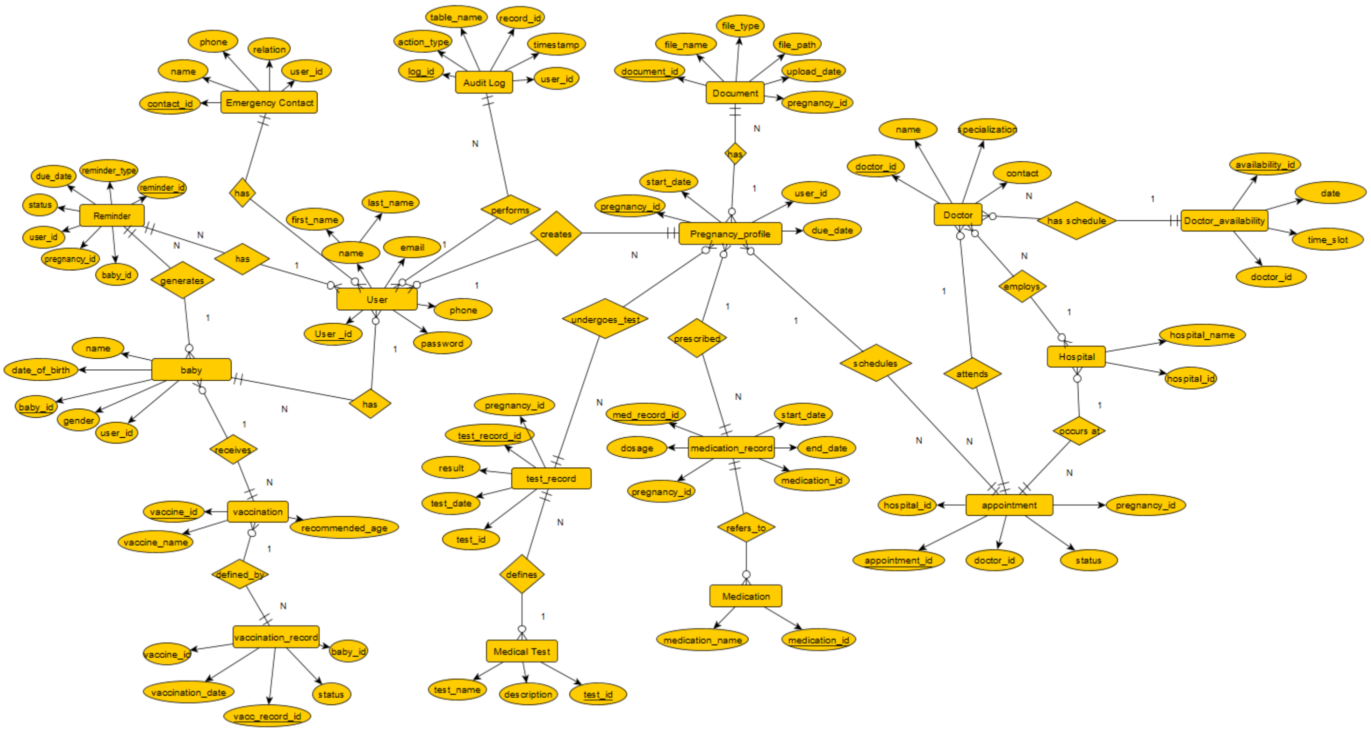
9. Document Management Module

This module stores pregnancy-related documents such as medical reports. It maintains details including file name, file type, file path, upload date, and associated pregnancy profile.

10. Audit Log Module

This module records system activities for monitoring and accountability. It stores information such as action type, affected table name, record ID, timestamp, and the user who performed the action. This module enhances system transparency and traceability.

ER Diagram



ER Diagram Constraints and Design Specifications

1. Entities

The MaterCare system consists of the following main entities:

Core Entities

- User
- Pregnancy_Profile
- Baby
- Doctor
- Hospital

Medical Tracking Entities

- Medical_Test
- Test_Record
- Medication
- Medication_Record
- Vaccination
- Vaccination_Record
- Appointment

System Support Entities

- Reminder
- Document
- Audit_Log
- Emergency_Contact

Each entity has a unique primary key used to identify records.

2. Key Constraints

Primary Keys (PK)

Every entity contains a primary key attribute ensuring uniqueness.

Examples:

- user_id → User
- pregnancy_id → Pregnancy_Profile
- baby_id → Baby
- appointment_id → Appointment
- log_id → Audit_Log

- reminder_id → Reminder

Primary keys guarantee entity integrity by preventing duplicate records.

Foreign Keys (FK)

Foreign keys establish relationships between entities and maintain referential consistency.

Examples:

- user_id in Baby references User.
- pregnancy_id in Appointment references Pregnancy_Profile.
- doctor_id in Appointment references Doctor.
- baby_id in Vaccination_Record references Baby.

Foreign keys enforce referential integrity.

3. Entity Integrity

Entity integrity ensures that:

- Each entity instance is uniquely identifiable.
- Primary key attributes cannot be NULL.
- All records are independent and valid.

For example:

- A Pregnancy_Profile cannot exist without a valid pregnancy_id.
- A Reminder must have a valid reminder_id.

4. Referential Integrity

Referential integrity ensures that:

- Child entities cannot reference non-existent parent entities.
- Relationships remain consistent across the database.

Examples:

- A Test_Record must reference an existing Medical_Test.
- A Medication_Record must reference a valid Pregnancy_Profile.

- A Vaccination_Record must reference an existing Baby.

5. Cardinality Constraints

Cardinality defines how many instances of one entity relate to another.

One-to-Many (1:N) Relationships

- User → Pregnancy_Profile
- User → Baby
- Pregnancy_Profile → Test_Record
- Pregnancy_Profile → Medication_Record
- Baby → Vaccination_Record
- Doctor → Appointment
- Hospital → Appointment
- User → Audit_Log
- User → Emergency_Contact
- User → Reminder

Optional Relationships

Some entities may exist without related records:

- Pregnancy_Profile may exist without Test_Record initially.
- Baby may exist without Vaccination_Record.
- User may exist without Reminder or Document.

6. Participation Constraints

Total Participation

The following entities depend on parent entities for existence:

- Appointment
- Test_Record
- Medication_Record
- Vaccination_Record
- Reminder

- Document
- Audit_Log
- Emergency_Contact

These entities cannot exist independently.

Partial Participation

The following entities may exist independently:

- User
- Pregnancy_Profile
- Baby
- Doctor
- Hospital
- Medical_Test
- Medication
- Vaccination

7. Associative Entities

Entities such as:

- Appointment
- Test_Record
- Medication_Record
- Vaccination_Record
- Reminder

act as associative entities that resolve many-to-many relationships while storing additional attributes.

8. Weak Entities

No strict weak entities are used in this ER model.
All entities maintain independent primary keys, even if they depend logically on parent entities.

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9. Design Assumptions

- Each doctor belongs to one hospital.
- Each pregnancy profile belongs to one user.
- Reminders may relate to pregnancy or baby data.
- The system focuses on data management rather than medical diagnosis.

Relational Schema